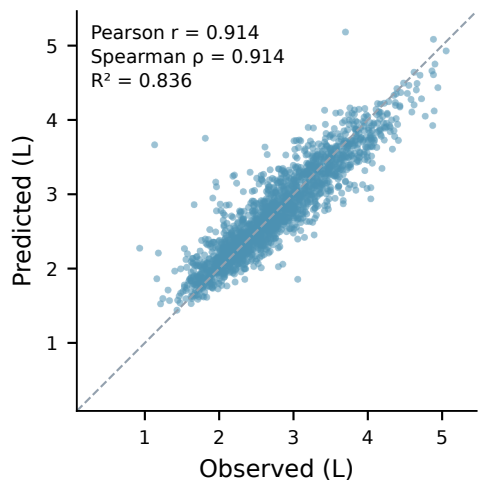
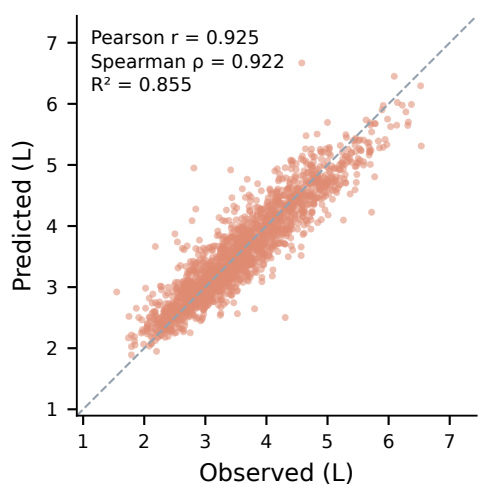
Healthy and COPD | performance and model interpretation

HEALTHY · fixed test, n = 1,899

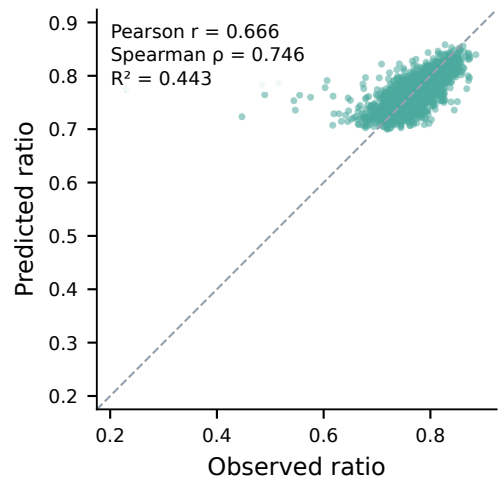
A1 FEV₁



A2 FVC

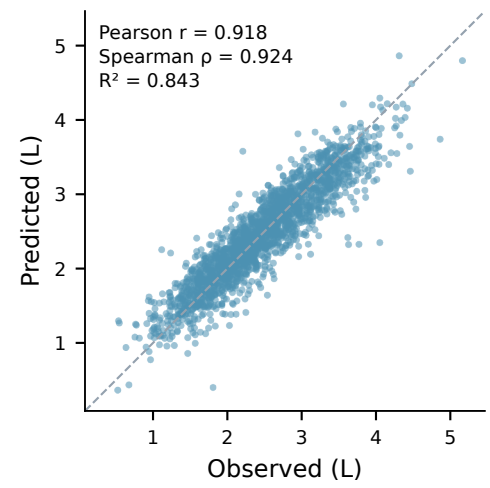


A3 FEV₁/FVC

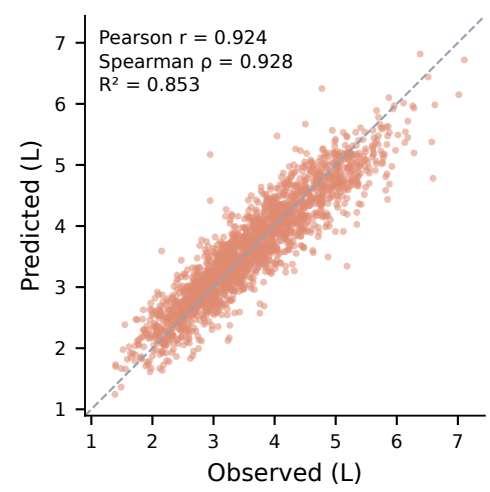


COPD · LOOCV, n = 2,067

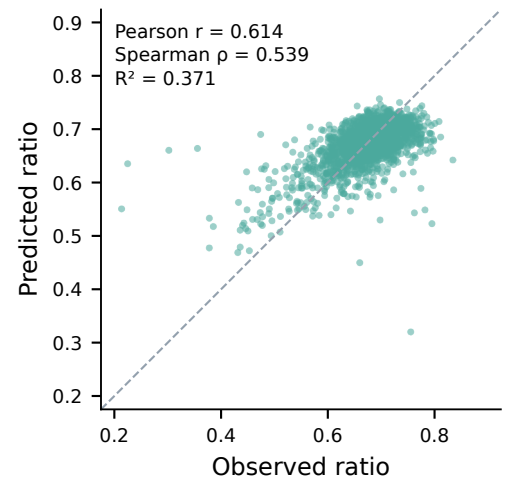
B1 FEV₁



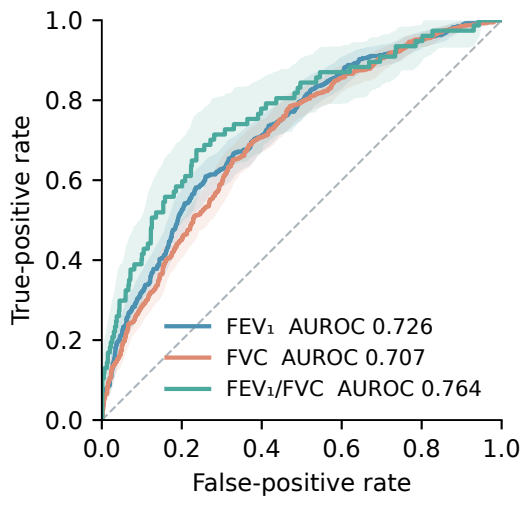
B2 FVC



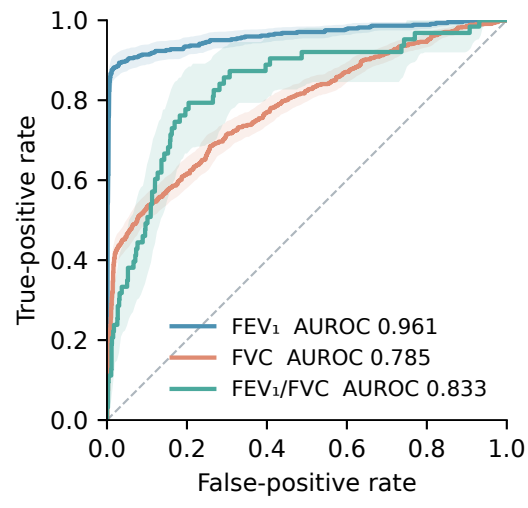
B3 FEV₁/FVC



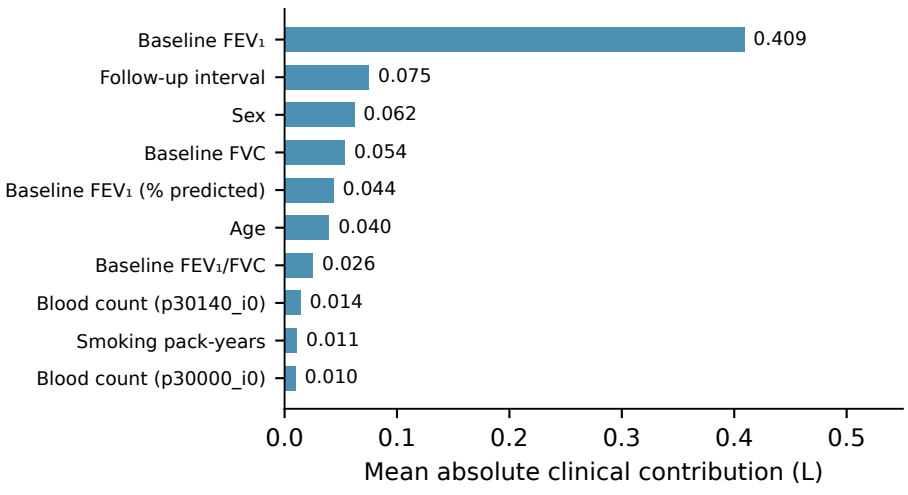
C Three rapid-decline ROC curves



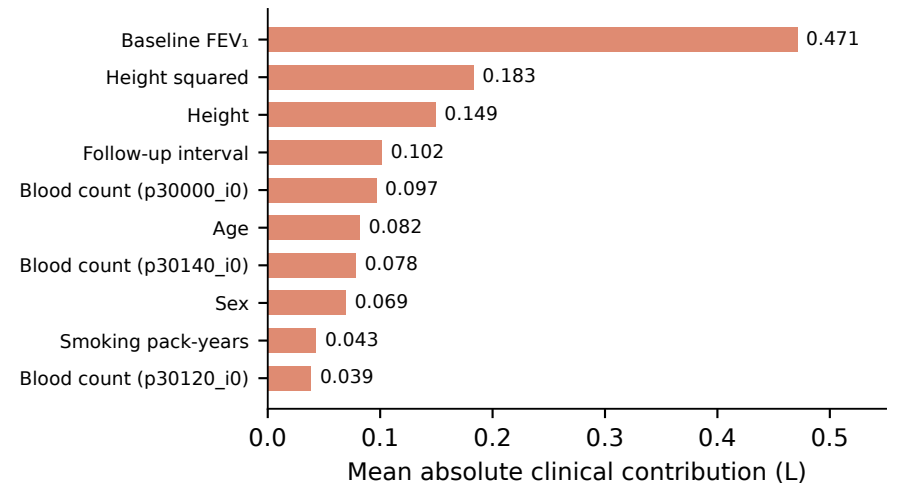
D Three rapid-decline ROC curves



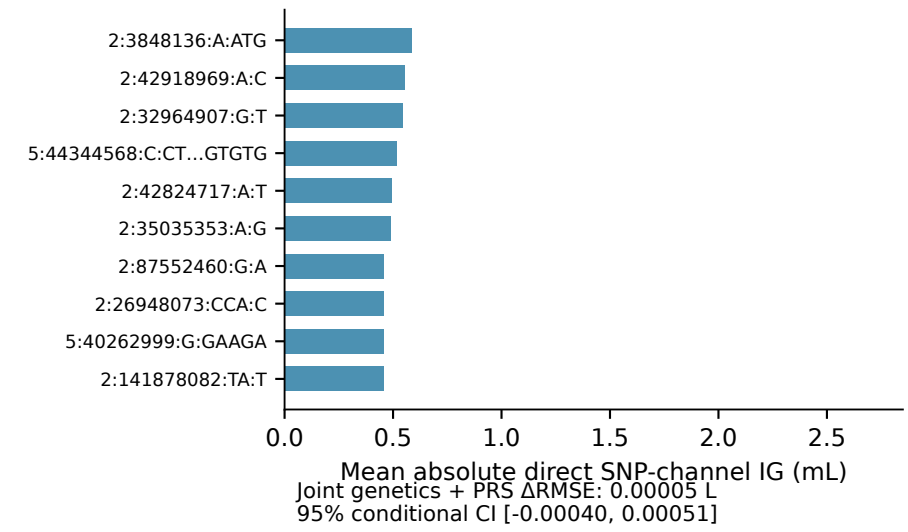
E Clinical Top 10 | next FEV₁



F Clinical Top 10 | next FEV₁

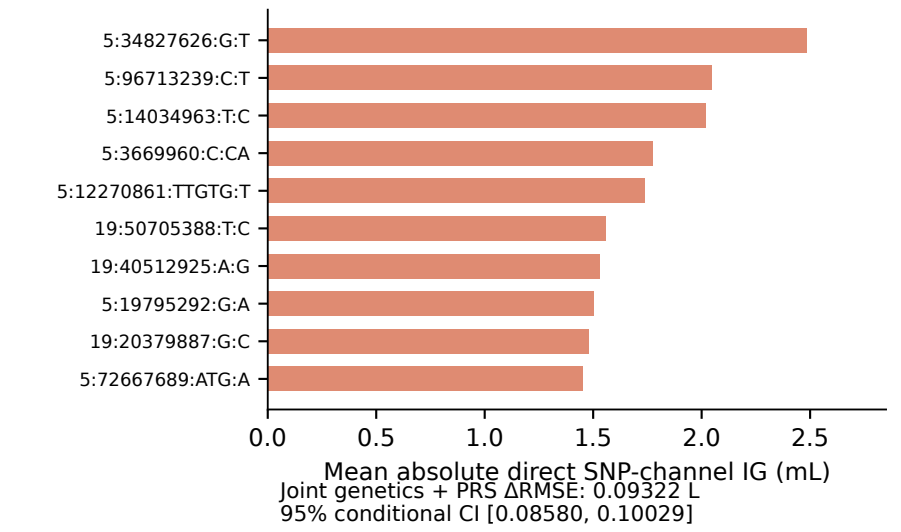


G Direct SNP-channel IG | next FEV₁



Joint genetics + PRS ΔRMSE: 0.00005 L
95% conditional CI [-0.00040, 0.00051]

H Direct SNP-channel IG | next FEV₁



Joint genetics + PRS ΔRMSE: 0.09322 L
95% conditional CI [0.08580, 0.10029]

Task colors: FEV₁ (blue), FVC (coral), FEV₁/FVC (teal). A1–A3 and B1–B3: separate tasks in original units; matched axes by task.

C–D bands: 500 patient bootstraps. E–F: native clinical attribution Top 10; Healthy linear + neural IG, COPD affine branch.

G–H show direct SNP input-channel IG only; PRS and clinical-SNP paths excluded. Different designs / inputs limit specificity claims.